



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

ENEE4113 - COMMUNICATION LAB

Experiment #1 PRELAB

Normal Amplitude Modulation and Demodulation

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Section: 4

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- **Theoretical Part:**

In an amplitude modulation (AM) communications system, a device is used to convert data into an electrical signal, for example, a microphone is used to convert audio into an electrical signal. This signal, known as the message or baseband signal, is then used to modify (modulate) the amplitude of another signal, known as the carrier signal.

The carrier wave (sine wave) of frequency f_c and amplitude A is expressed by

$$c(t) = A \sin(2\pi f_c t) .$$

The message signal, such as an audio signal that is used for modulating the carrier, is $m(t)$, and has a frequency f_m , much lower than f_c :

$$m(t) = M \cos(2\pi f_m t + \phi) = Am \cos(2\pi f_m t + \phi) ,$$

where m is the amplitude sensitivity, M is the amplitude of modulation. If $m < 1$, $(1 + m(t)/A)$ is always positive for undermodulation. If $m > 1$ then overmodulation occurs and reconstruction of message signal from the transmitted signal would lead in loss of original signal. Amplitude modulation results when the carrier $c(t)$ is multiplied by the positive quantity $(1 + m(t)/A)$:

$$\begin{aligned} y(t) &= \left[1 + \frac{m(t)}{A} \right] c(t) \\ &= [1 + m \cos(2\pi f_m t + \phi)] A \sin(2\pi f_c t) \end{aligned}$$

Using prosthaphaeresis identities, $y(t)$ can be shown to be the sum of three sine waves:

$$y(t) = A \sin(2\pi f_c t) + \frac{1}{2} Am [\sin(2\pi [f_c + f_m] t + \phi) + \sin(2\pi [f_c - f_m] t - \phi)] .$$

Therefore, the modulated signal has three components: the carrier wave $c(t)$ which is unchanged in frequency, and two sidebands with frequencies slightly above and below the carrier frequency f_c .

- **Software Part:**

- **Block Simulation (Matlab Simulink):**

The modulation :

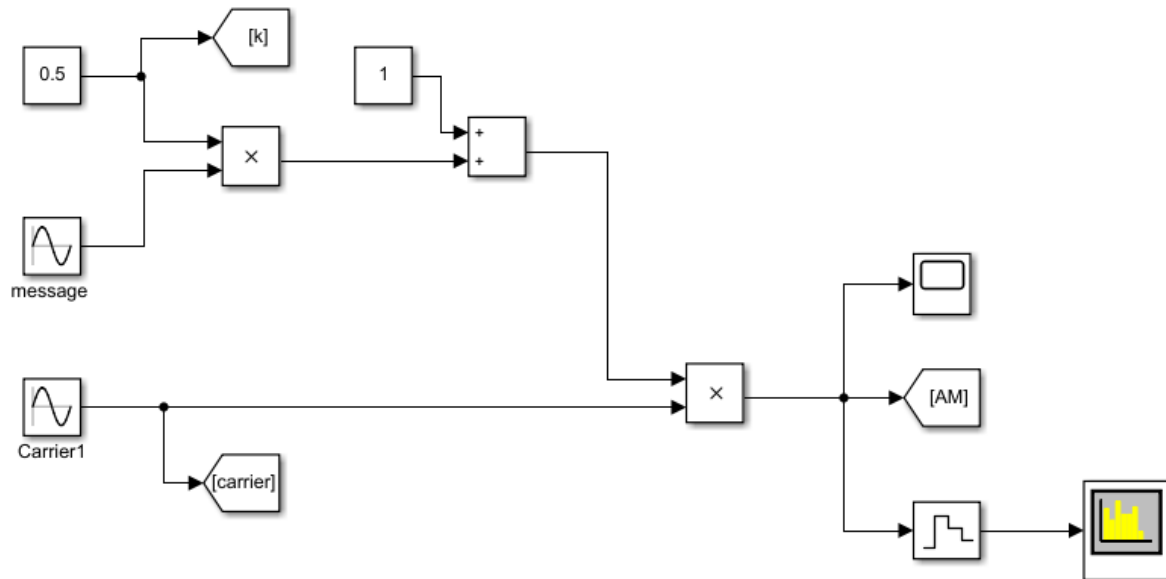


Figure 1:NA modulation block simulation

The demodulation :

1. Coherent Demodulation :

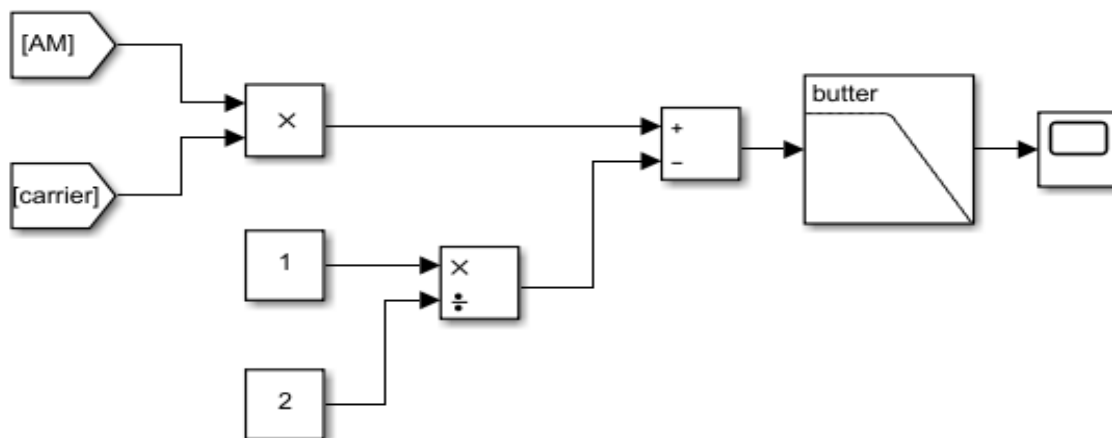


Figure 2:Coherent Demodulation block simulation

2. Envelope Detector Demodulation :

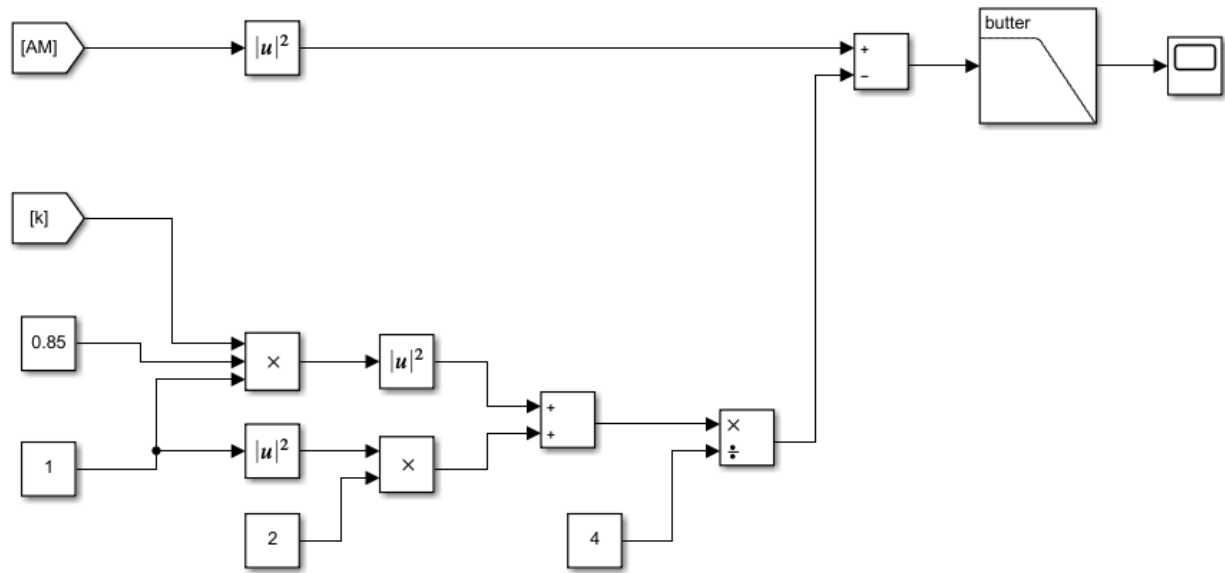
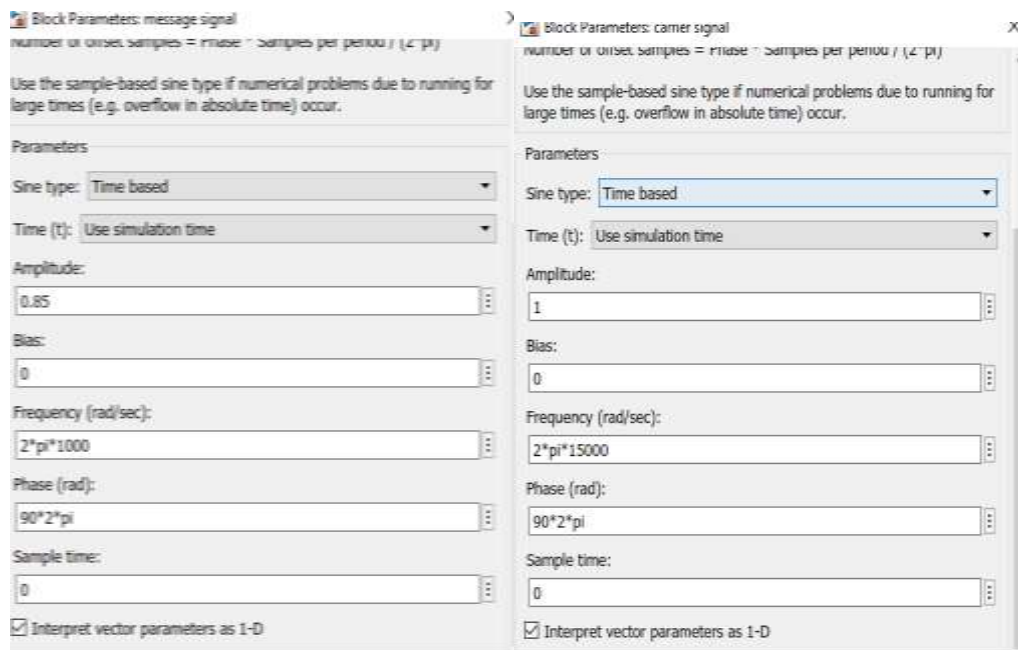


Figure 3:Envelope Detector Demodulation block simulation

➤ The sitting :



➤ **The Results :**

1. The modulation :

➤ At $M = 0.5$

• In frequency domain :

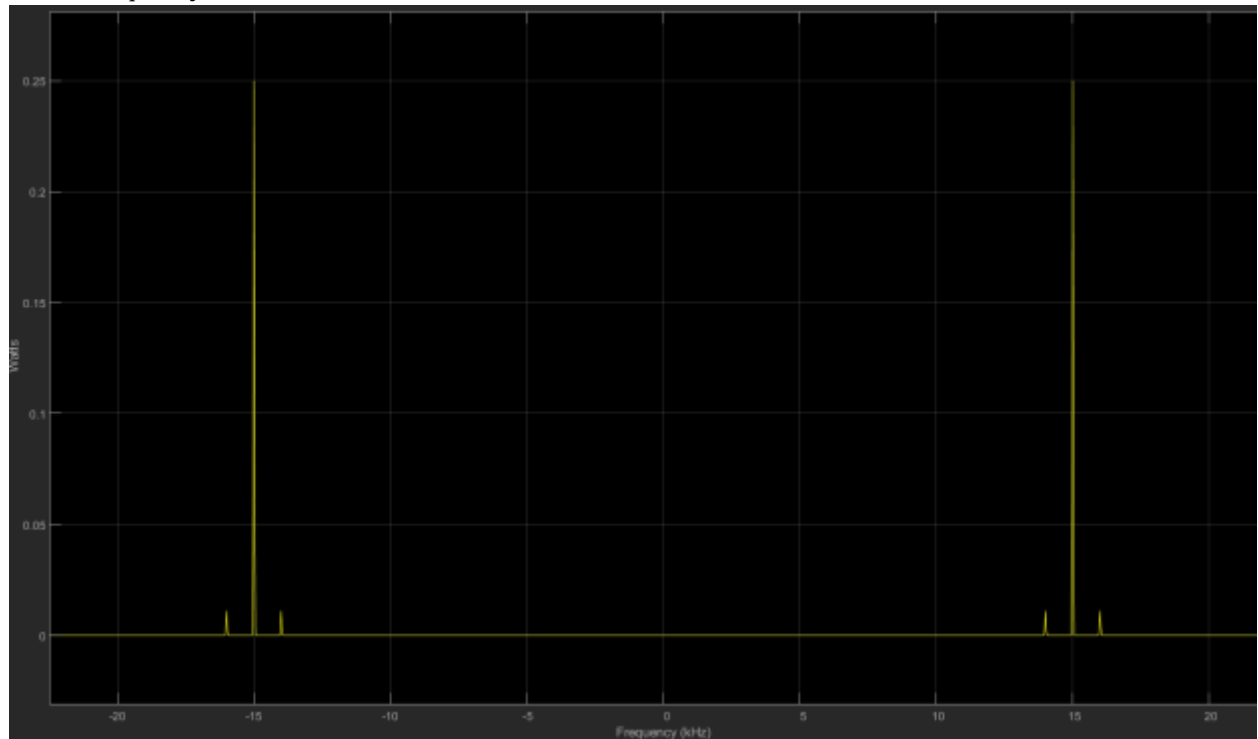


Figure 4:AM modulated signal at 0.5

• In time domain:

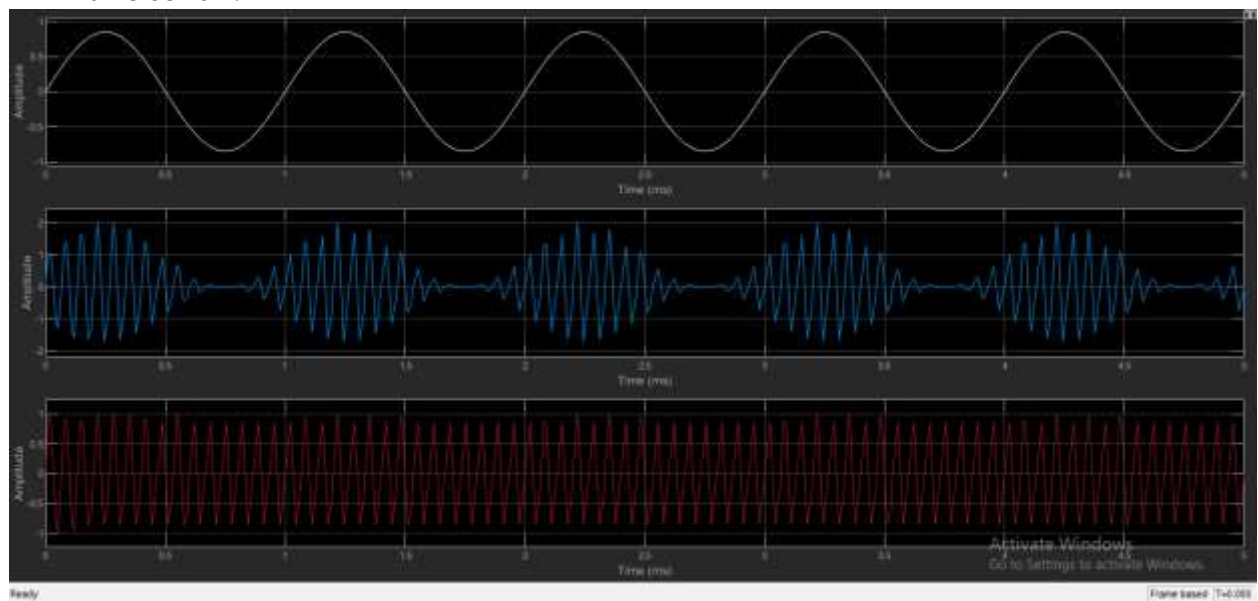


Figure 5:NA in time domain at 0.5

- At $M=1$:
- In frequency domain :

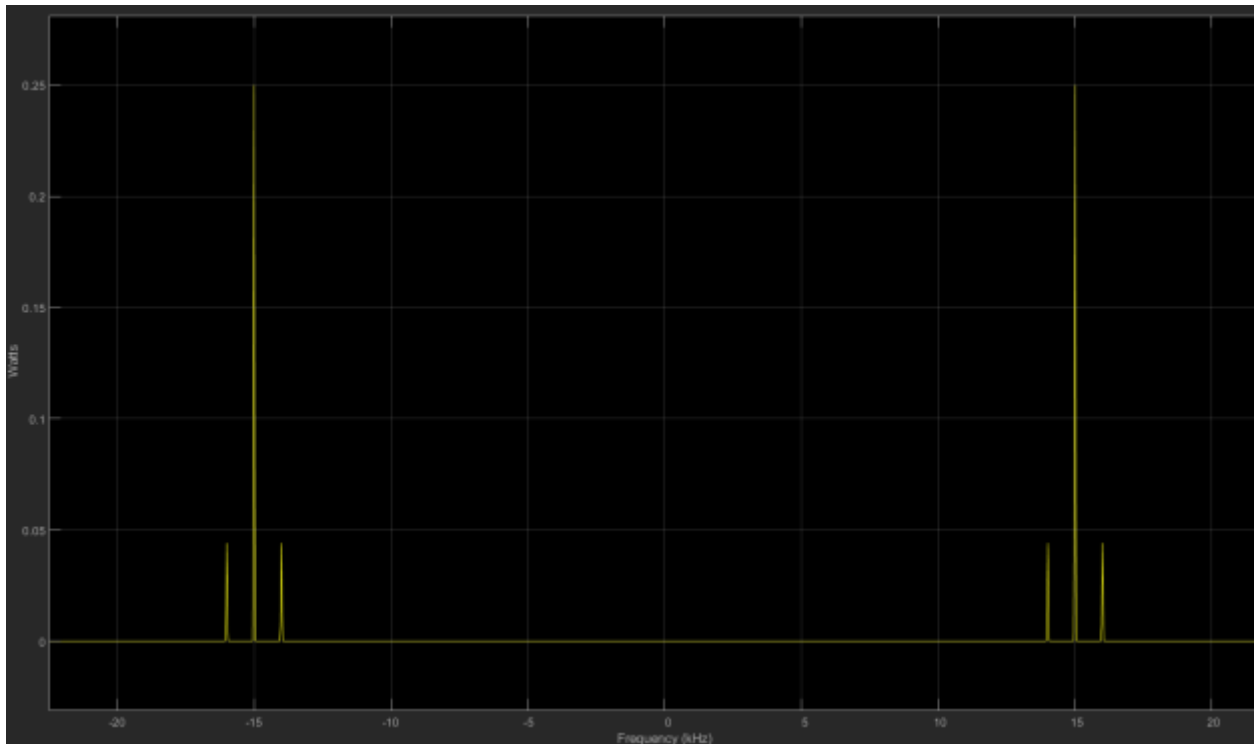


Figure 6: AM modulated signal at 1

- In time domain :

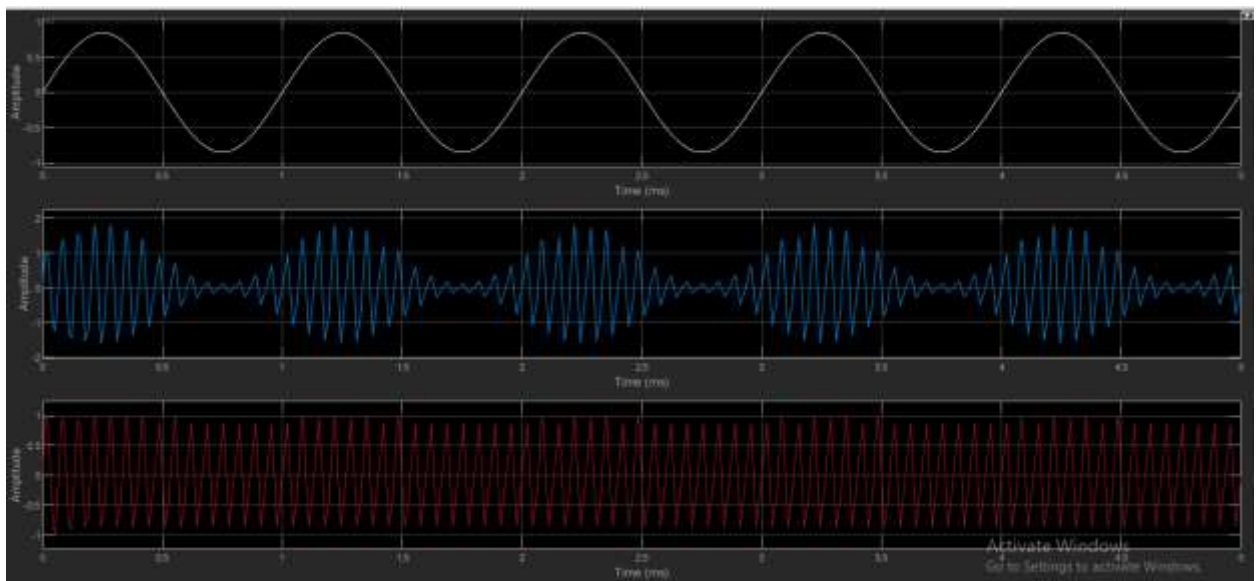


Figure 7: AM modulated signal at 1 in time domain

- At $M = 3$:
- In frequency domain :

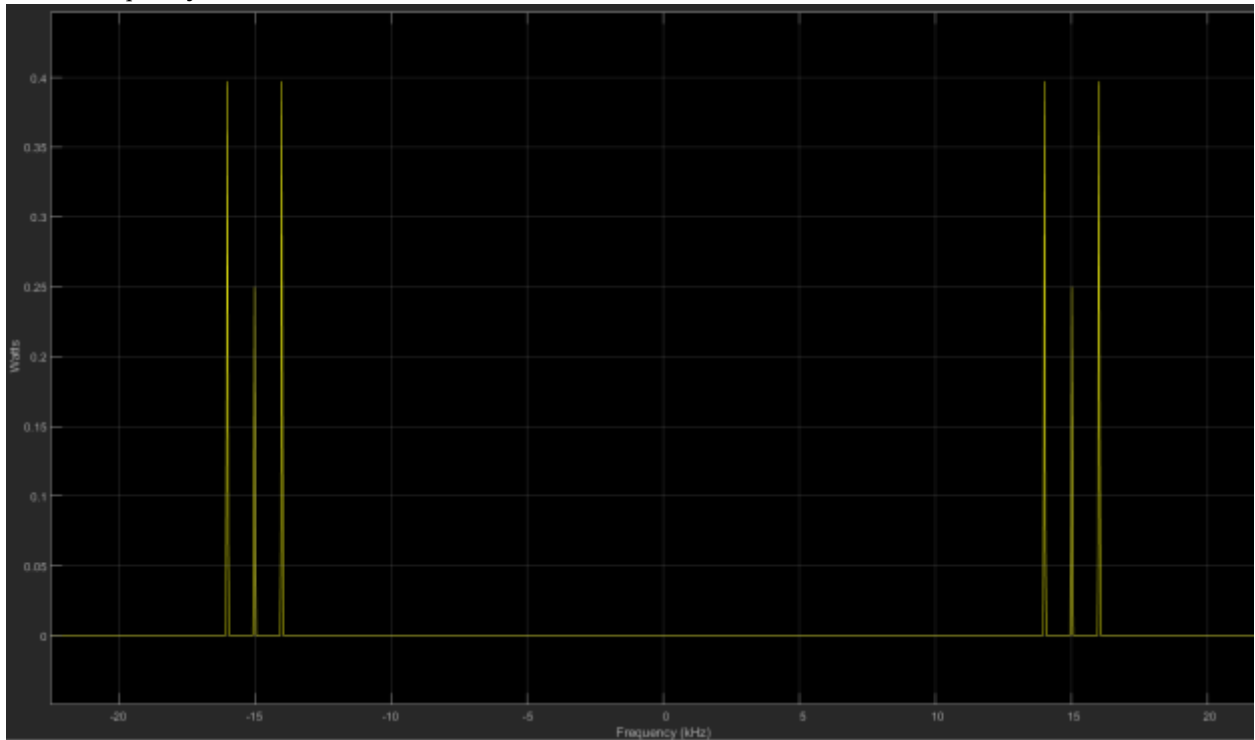


Figure 8:AM modulated at 3

- In time domain:

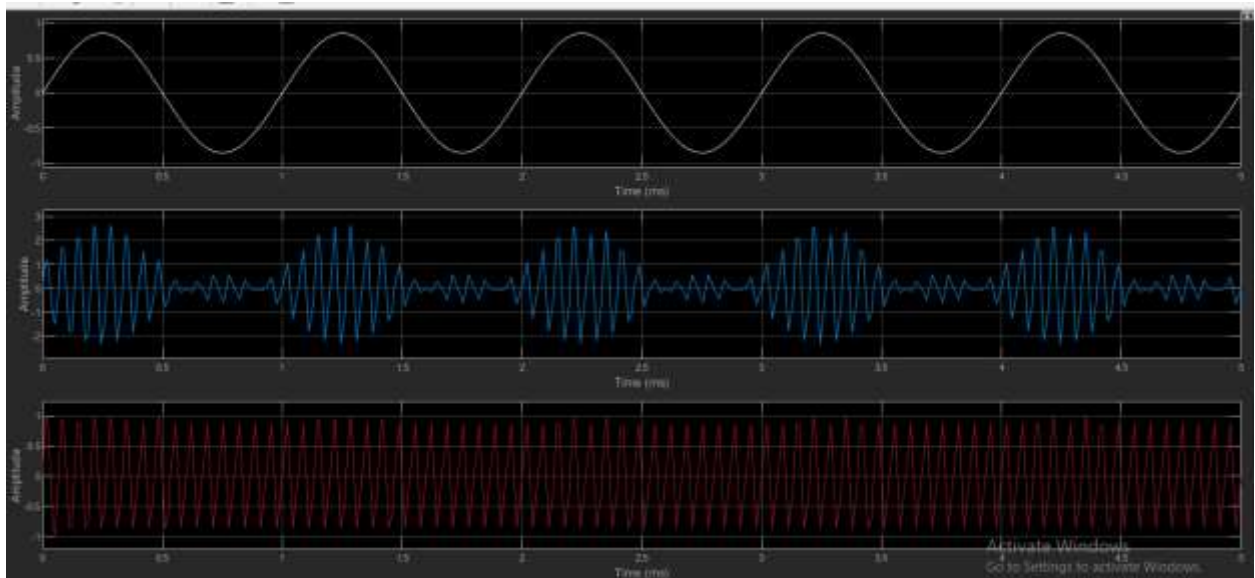


Figure 9 : at 3 in time domain

2. The demodulation :

- At $M = 0.5$:
- In frequency domain :

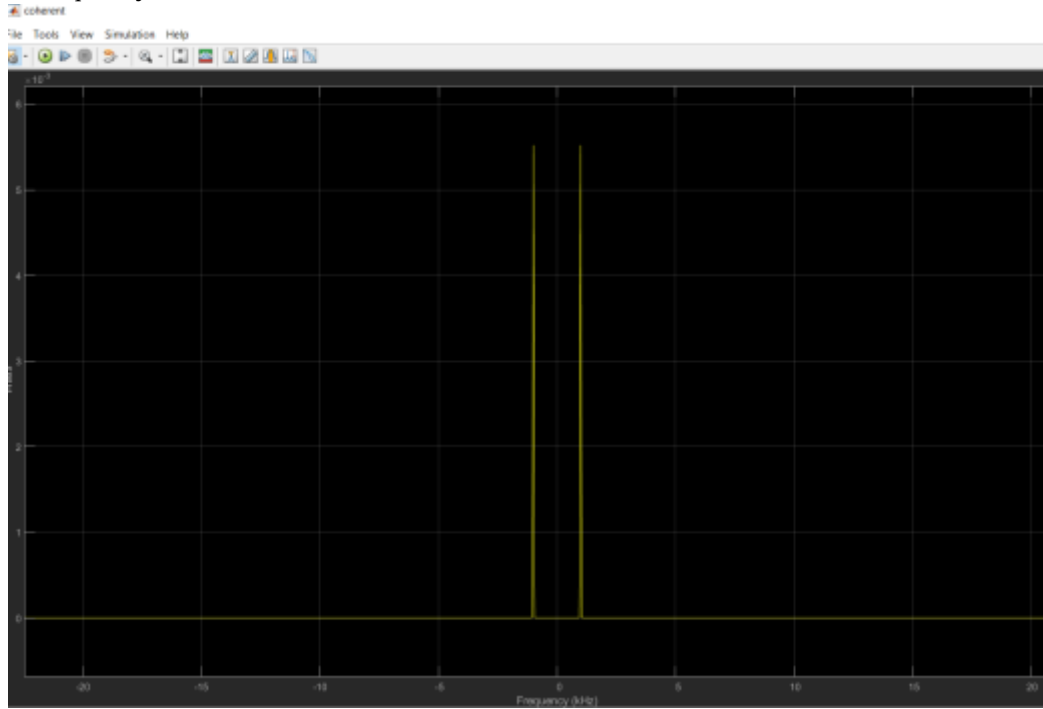


Figure 10:coherent demodulation at 0.5

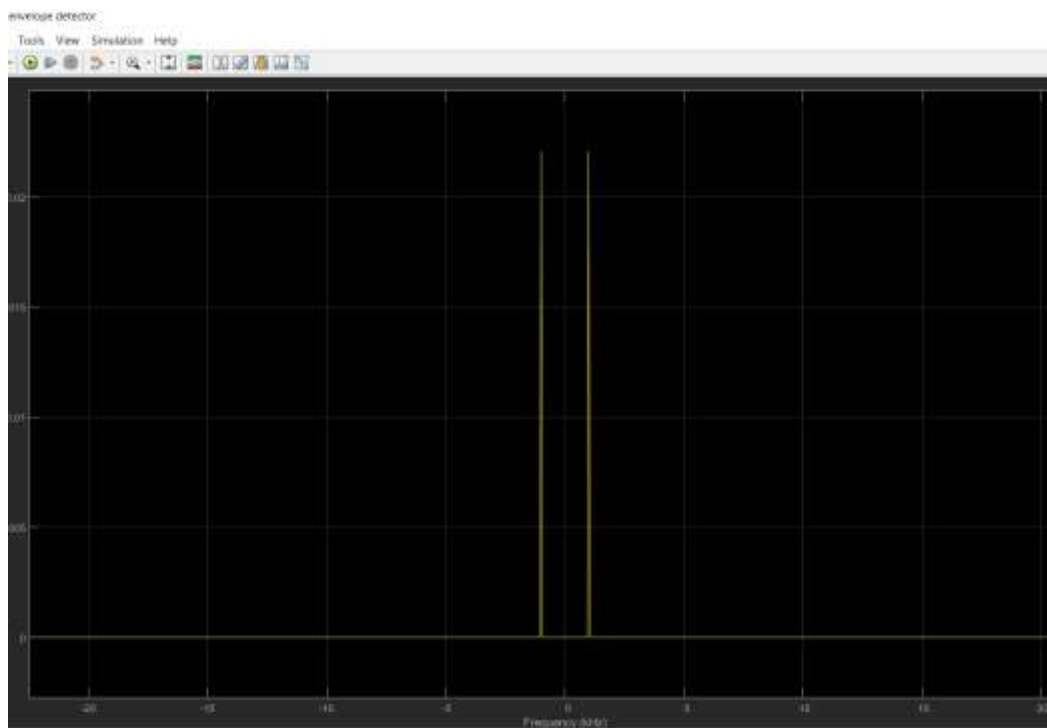


Figure 11: envelope detector demodulation at 0.5

- In time domain:

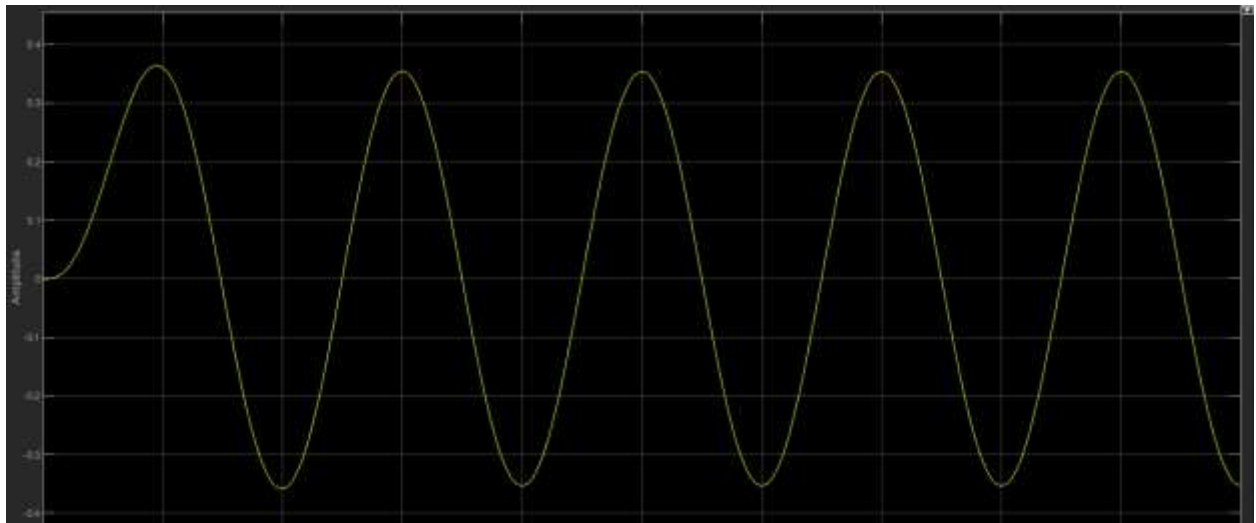


Figure 12 : demodulated at 0.5 in time domain

- At $M = 1$:
- In frequency domain :

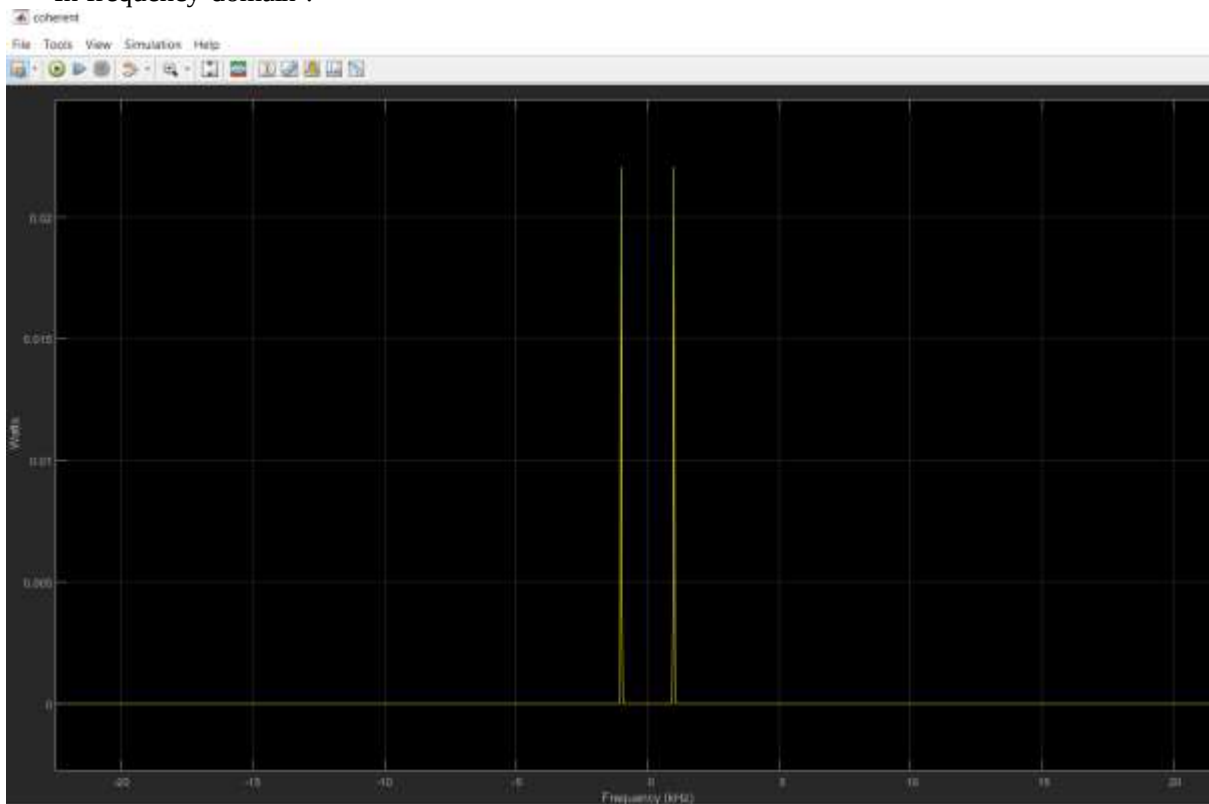


Figure 13:coherent demodulation at 1

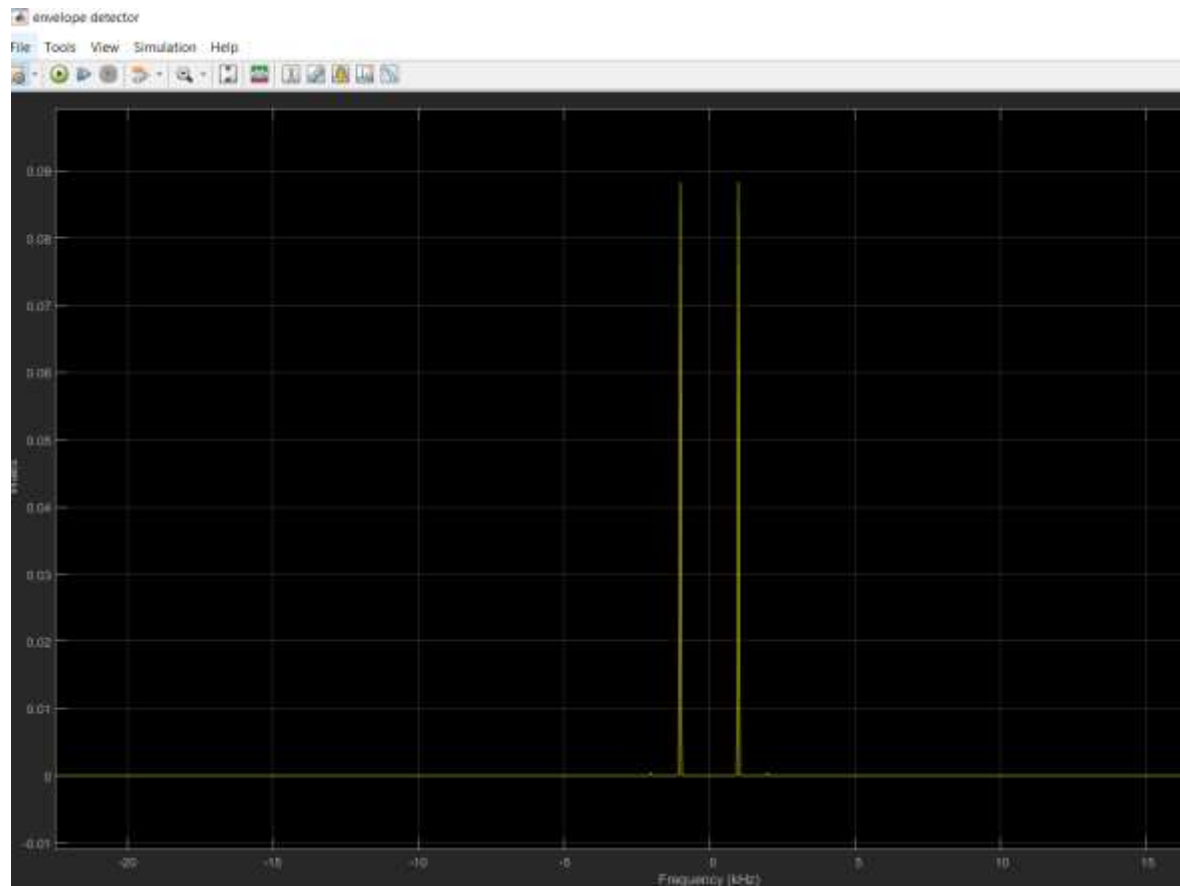


Figure 14: envelope detector demodulation at 1

- In time domain :



Figure 15: demodulated at 1 in time domain

➤ At $M = 3$:

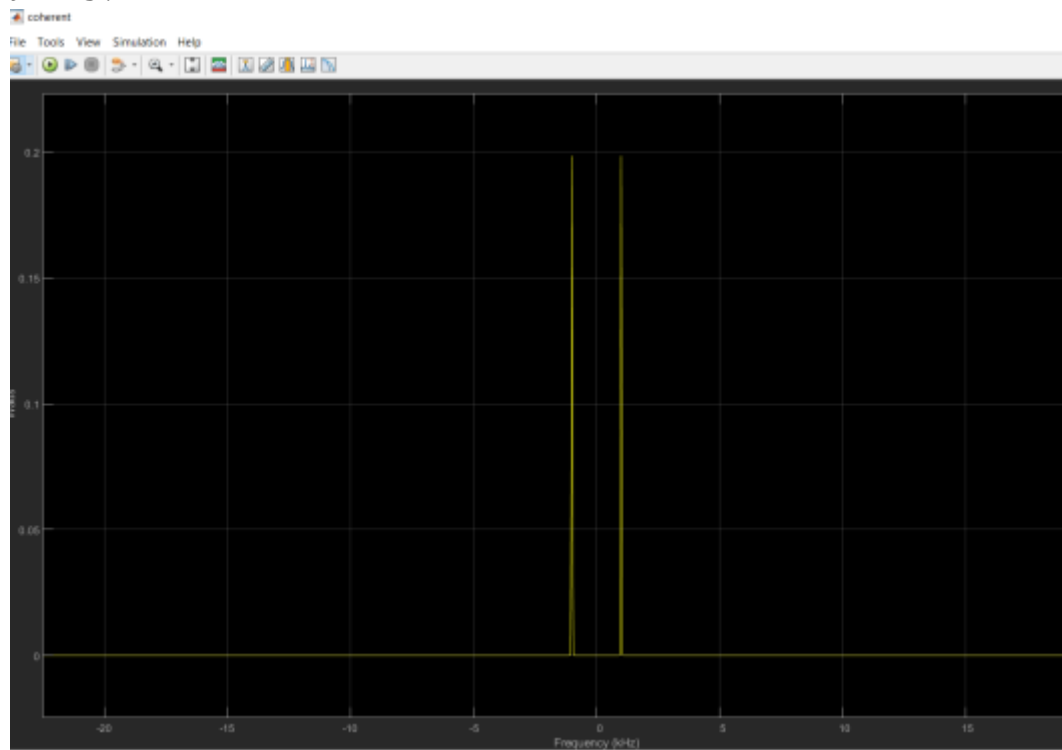


Figure 16: coherent demodulation at 3

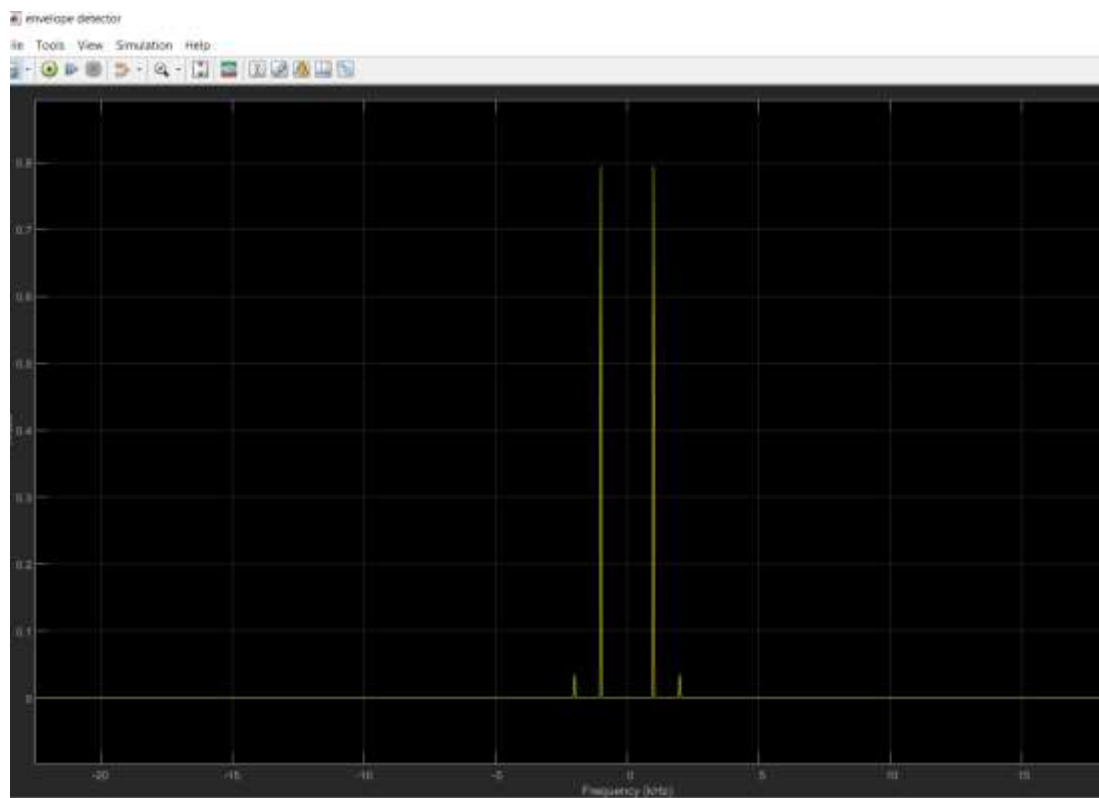


Figure 17: envelope detector demodulation at 1

- In time domain :



Figure 18: demodulated at 3 in time domain